

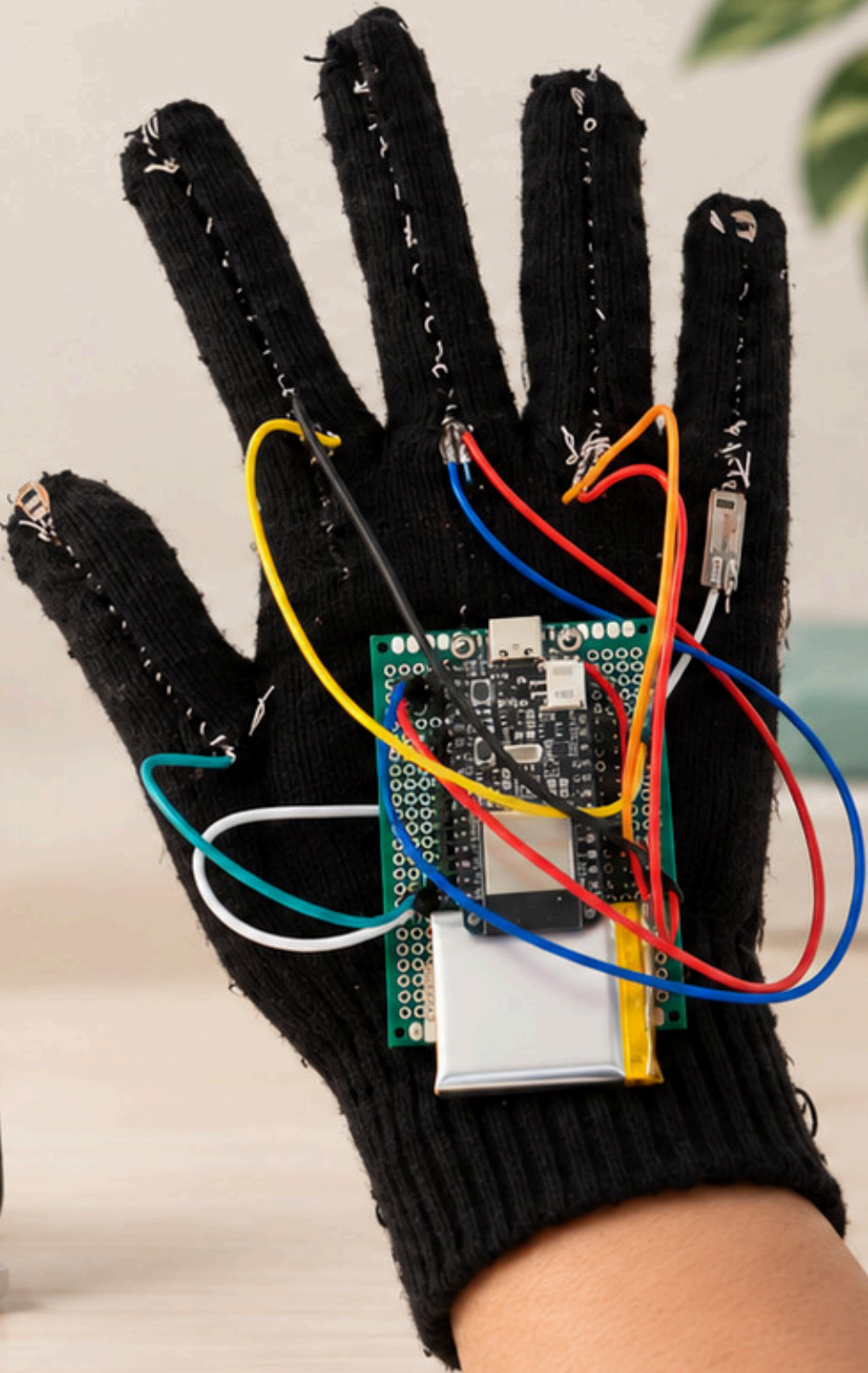
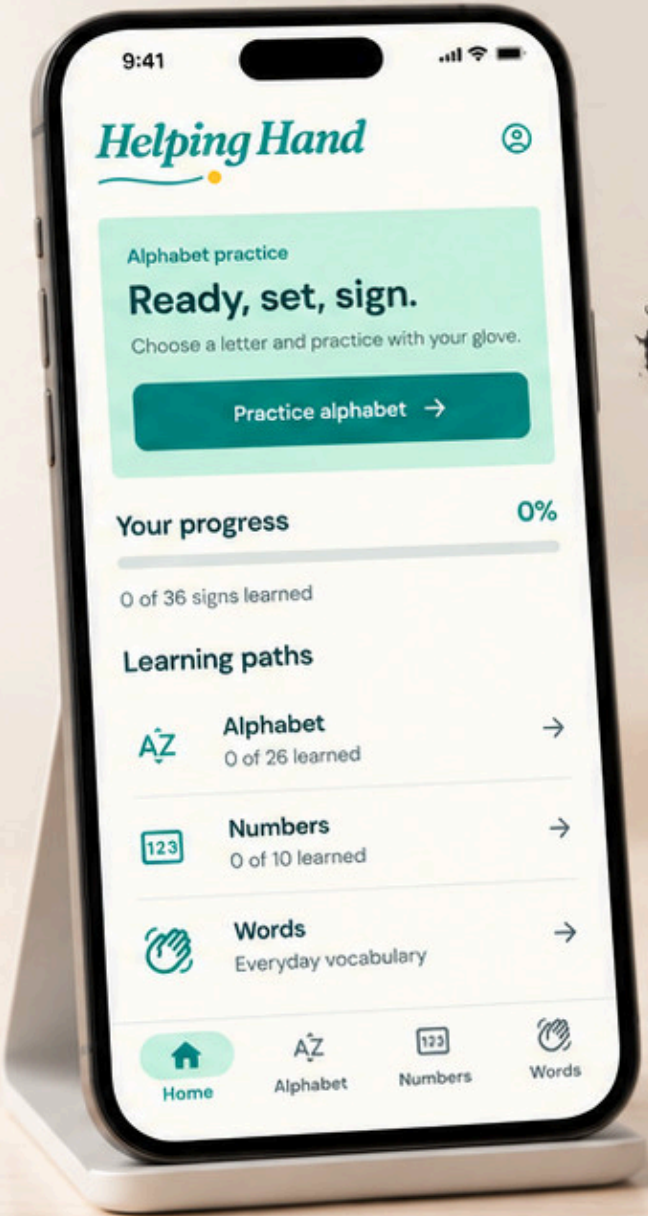
BETA BUILD · OCTOBER 6, 2026

Helping Hand

A Wearable ASL Learning System

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THE PROBLEM

Learning ASL needs feedback.

>90%

of deaf children are born to hearing parents.

Books and videos show the sign.

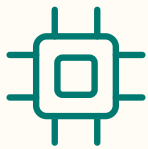
2D examples cannot assess your hand position.

No live correction from the reference alone.



OUR SOLUTION

Wear. Sign. Get feedback.



Sensor glove

Flex + IMU

ESP32

Capture readings

BLE

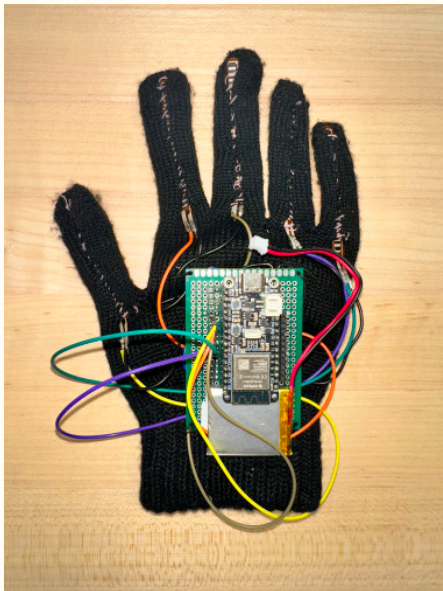
Stream packets

Flutter app

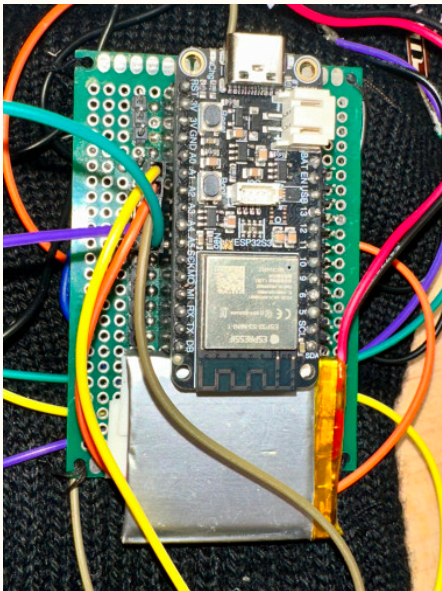
Recognize signs

Feedback

Guide practice



Glove · flex sensors + mounted electronics



Perfboard · ESP32-S3 + sensor wiring

The first complete pipeline



Alpha glove



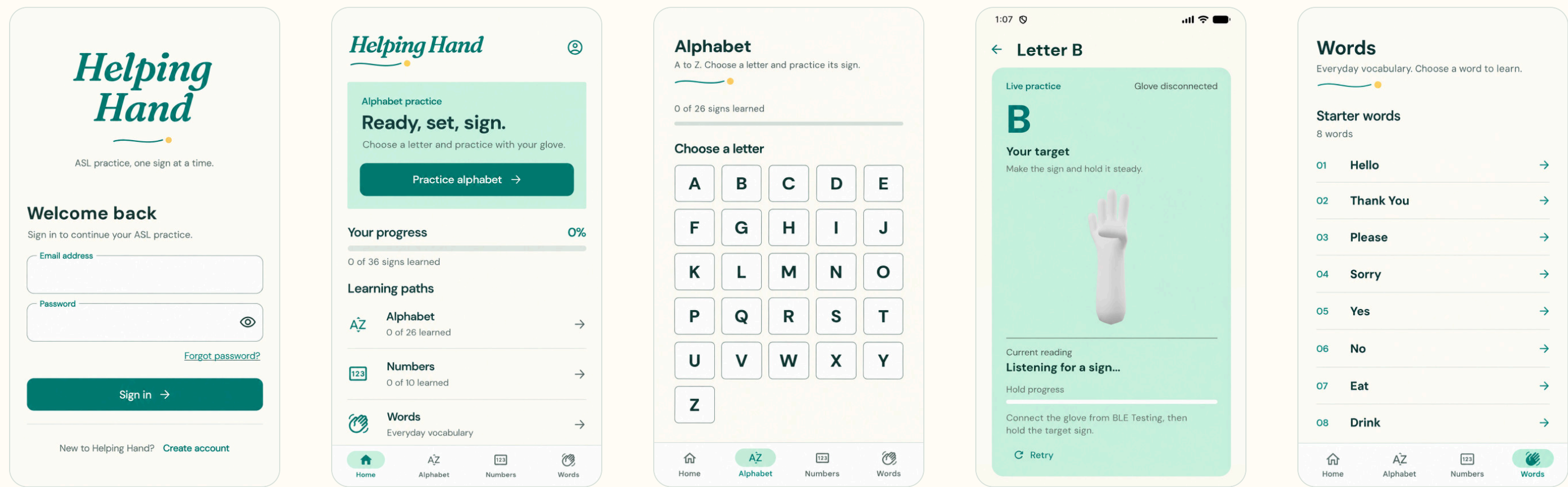
Alpha interface

- 01 **Sensor capture**
Finger bend + motion readings
- 02 **Wireless streaming**
Glove → BLE → app
- 03 **Static classification**
36-class TFLite baseline

THE BETA APP

A clearer path to practice

Firebase sign-in · dedicated practice pages · account-specific progress



01 Sign in

02 Choose a path

03 Pick a target

04 Practice

05 Explore words

Recognizing words with a TCN

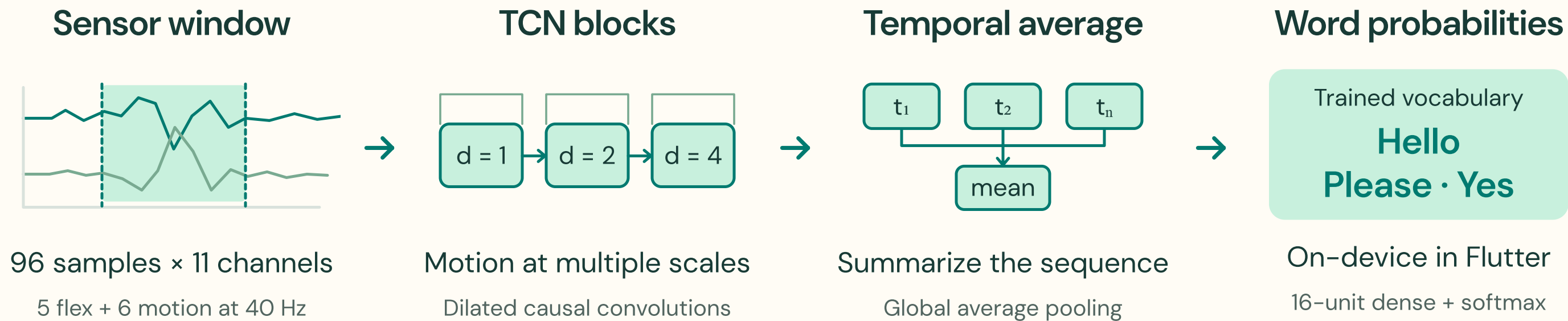
Static · MLP

One sensor snapshot → static class

Words · Temporal Convolutional Network

Sensor window → word prediction

Resample, trim or pad, and standardize before inference



Compared with CNN-GRU and CNN-LSTM

HELPING HAND IN ACTION

LIVE DEMO!!!

Static sign → Word attempts

Implementation and remaining validation

Completed implementation

Core experience

- Static MLP + 3-word TCN
- Firebase accounts + lessons
- Saved sign + word progress

In progress

Live validation

- Glove + phone rehearsal
- Rest / unknown rejection
- Wrist angles + other wearers

Planned

Tactile guidance

- 10 planned piezo elements
- Error-to-vibration mapping
- Calibration + user testing

Static completion rules

$\geq 80\%$

Confidence

$\geq 750\text{ ms}$

Stable matching hold

≥ 5

Matching packets

$\leq 250\text{ ms}$

Gap between packets

NEXT MILESTONE

Feel what needs correcting.

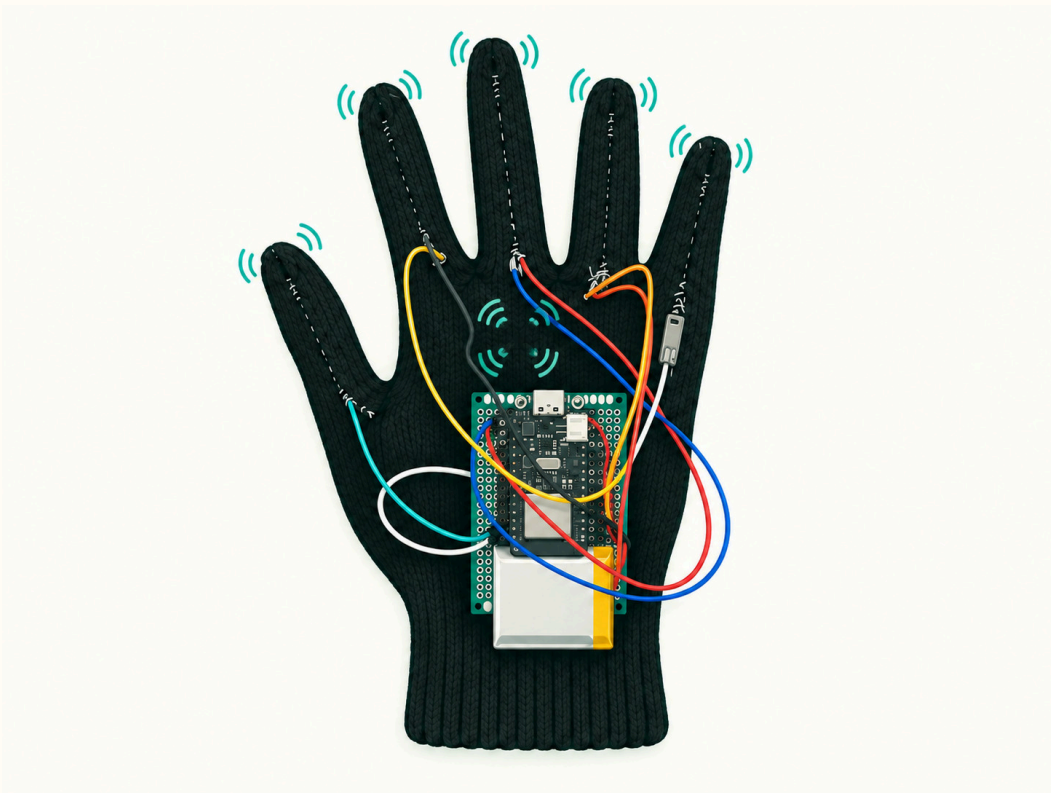
5 + 5

X pattern + one on each finger

Planned piezo feedback for incorrect positioning.

Estimated deviation → Pulse pattern / intensity

Placement, error mapping, and tactile response need calibration.



Concept illustration · planned feature

Oct 7–13

Mount + wire

Actuators and drivers

Oct 14–20

Calibrate guidance

Error → tactile response

Oct 21–27

Refine recognition

Orientation + signer trials

Oct 28–Nov 3

Test + polish

User trials and release prep

Helping Hand

Thank you. Questions?

Practice with feedback.

Next: guidance you can feel.